Module 3 Assignment — Computational Lab 3

Reranking. I combined the disease-relevance evidence from Module 2 with the sequence-based tractability prior from Module 3. I used a weight of 0.6, so the Module 2 disease evidence contributed 60% and tractability prior contributed 40% to the final score. I used this w because disease relevance should remain the main factor, while tractability should still influence target prioritization. Overall, the Module 3 ranking was broadly consistent with the Module 2 ranking, but some targets changed position.

From the targets_ov week 3 table, for the targets which moved most, BCHE moved up the most, which is from rank 13 to rank 8, because it had a relatively high small molecule tractability prior (p_sm = 0.738). In contrast, DMD (from rank 3 to rank 7) and TTC3 (from rank 8 to rank 12) each dropped four positions because both had very small molecule and antibody tractability scores. These results show that a target with strong disease evidence does not necessarily remain highly ranked once potential druggability is considered.

Modality. For small molecule versus antibody tractability, I would choose BCHE that showed a clear difference between the two modality priors, with p_sm = 0.738 and p_ab = 0.030. This suggests that BCHE is more similar to proteins that have historically been tractable with small molecules than to proteins commonly targeted by antibodies. This difference may reflect biological features such as protein family & fold, domain composition, and localization & topology, which can influence therapeutic modality. The higher small-molecule prior also helps explain why BCHE moved upward in the final re-ranking.

However, these scores are only sequence-based tractability priors and should not be interpreted as proof that BCHE has a true druggable pocket or that a small-molecule drug will

succeed. Structural analysis is still needed to determine whether an appropriate binding pocket is actually present.

Validation. The family-spit metrics achieved an AUROC of 0.732 and an average precision (AP) of 0.542. These values were lower than the random-split results, which were 0.834 AUROC and 0.709 average precision. I would report the family-split metrics results because I think it is closer to the realistic estimate of model performance. In a family split, proteins from the same family are kept together, so closely related proteins do not appear in both the training and test sets. This makes the evaluation more difficult, but it better reflects the real situation in which the model may need to score a target from a protein family it has not seen before.

In contrast, a random split can place highly similar or homologous proteins on both sides of the split. This creates homology leakage, because the model may be tested on proteins that are very similar to proteins it already saw during training. Therefore, performance based on a random split may appear better than the model's true generalization capability. The drop from 0.834 to 0.732 in AUROC and from 0.709 to 0.542 in average precision shows that some of the apparent performance under random splitting likely came from this similarity between training and test proteins. The family-split average precision of 0.542 is still above the chance level of 0.334, suggesting that the model retains useful predictive signals even when whole protein families are held out.

Variants. The four anchor variants showed different protein-language-model scores. APOE R176C (ε2) had the most negative score, −8.56, with a within-protein percentile of 11.8, indicating that the model considers this substitution relatively poorly tolerated. TP53 R175H

also had a negative score of −5.97 with a percentile of 19.3, while TREM2 R47H had a score of −5.16 and a percentile of 47.7. In contrast, APOE C130R (ε4) had a positive score of 7.21 and a percentile of 100.0, suggesting that the model does not consider this substitution evolutionarily surprising.

A strongly negative score indicates that the model finds the substitution surprising and it does not necessarily mean that a variant is clinically pathogenic. The score mainly reflects how unusual or poorly tolerated a missense substitution appears based on the sequence patterns learned by the model. In addition, protein-sequence models cannot directly evaluate splicing, regulatory, or noncoding effects. Therefore, the variant scores should be interpreted as evidence of evolutionary tolerance rather than as a direct clinical diagnosis.

A sequence-based tractability prior can help prioritize targets, but it should be treated as a prior rather than proof of druggability. Protein language models learn evolutionary patterns and can identify proteins that resemble known drug targets, but they do not provide physical protein structure. They may capture information related to contacts, but they cannot directly show whether a real binding pocket exists or where it is located. The model can also be less reliable for proteins from poorly represented families, because those proteins have fewer close relatives in the training data. Therefore, the Module 3 results are most useful for target triage and for deciding which proteins should be examined more closely in the next step, rather than for making a final conclusion about druggability.

For Week 4, I would select BCHE and PGR for structure-based pocket assessment. BCHE had a high small-molecule prior (p_sm = 0.738) and showed the largest upward movement in the re-ranking, moving from rank 13 in Week 2 to rank 8 after tractability was included. This suggests that BCHE has a strong sequence-based tractability signal and is worth checking for

a real pocket. PGR remained the top-ranked target overall and had a moderate small-molecule prior (p_sm = 0.410) together with strong Week 2 evidence. Examining both targets in Week 4 would allow the sequence-based priorities from Module 3 to be compared with structure-based pocket evidence before deciding which targets should move forward.

All of my top 15 targets were successfully mapped to the reviewed human proteome.